Programming for Economist

PS 6 & 7

Class 3

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Plan for today

1. Replacement class
2. Numerical optimisation/root finding
3. Interpolation
4. Work on PS6/7

Let me know if you would like to go through some questions on the board:)

2. Numerical optimisation

- Uses an algorithm to find inputs that maximise/minimise (or finds a root to) a function, e.g. the consumption bundle that gives the highest utility.
- No longer constrained by analytical solutions, instead we are constrained by computing time.
  - Curse of dimensionality
- Keep in mind:
  - The result is usually an approximation, not an exact solution.
  - An algorithm may find a *local* optimum rather than the *global* optimum.
  - Results can depend on starting values, constraints, and the shape of the function.
  - Check whether the solver succeeded, whether the solution is feasible, and whether different starting values give similar answers.

3. Interpolation

- Interpolation estimates a function's value **between observed points**.
- Used when we do not know the function itself, but have values evaluated at two or more input points, e.g. a table of tax rates and corresponding tax bases.
- Simplest form is linear interpolation.
- Keep in mind:
  - Estimates depend on the interpolation method and the available data.
  - Interpolation does not reveal the true function; estimates may be inaccurate between widely spaced points.
  - Estimating beyond the observed points is **extrapolation**, which is generally less reliable.